# Question-12.36

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# Question

Given the regression model: $Y_i = \alpha + \beta x_i + \epsilon_i$ with data:

| Year | Production |
|---|---|
| 2001 | 30 |
| 2002 | 35 |
| 2003 | 36 |
| 2004 | 32 |
| 2005 | 37 |
| 2006 | 40 |

(1)

# Solution

Let

$$x_i = \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \end{pmatrix}, Y = \begin{pmatrix} 30 \\ 35 \\ 36 \\ 32 \\ 37 \\ 40 \end{pmatrix} \tag{2}$$

Let

$$X = \begin{pmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \\ 1 & 5 \\ 1 & 6 \end{pmatrix} \tag{3}$$

# Solution

$$X^TX = \begin{pmatrix} 6 & 21 \\ 21 & 91 \end{pmatrix}, X^TY = \begin{pmatrix} 210 \\ 761 \end{pmatrix} \tag{4}$$

$$\left(X^TX\right)^{-1} = \begin{pmatrix} 0.87 & -0.20 \\ -0.20 & 0.06 \end{pmatrix} \tag{5}$$

$$\begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \left(X^TX\right)^{-1}\left(X^TY\right) \tag{6}$$

$$= \begin{pmatrix} 0.87 & -0.20 \\ -0.20 & 0.06 \end{pmatrix} \tag{7}$$

$$= \begin{pmatrix} 210 \\ 761 \end{pmatrix} = \begin{pmatrix} 29.80 \\ 1.49 \end{pmatrix} \tag{8}$$

Thus, $\alpha + \beta = 31.29$

# graph

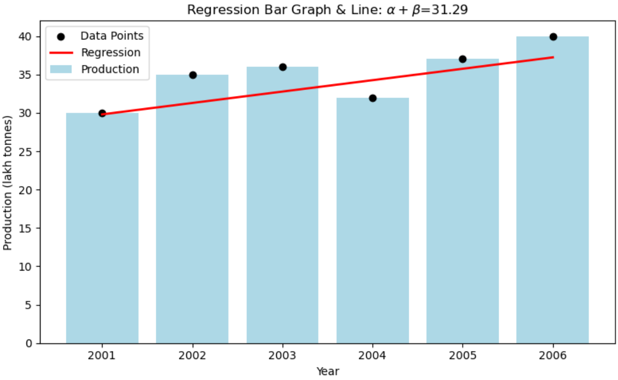


Figure:

# C-Code

```c
// code.c
#include <stdio.h>

void get_data(double *years, double *production, int *n) {
    // Fill arrays with dataset, n=6
    double local_years[6] = {2001, 2002, 2003, 2004, 2005, 2006};
    double local_prod[6] = {30, 35, 36, 32, 37, 40};
    for(int i=0; i<6; i++) {
        years[i] = local_years[i];
        production[i] = local_prod[i];
    }
    *n = 6;
}
```

# C-Code

```c
void get_stats(double *years, double *production, int n, double *out) {
    // Linear regression using normal equations, years mapped to 1,2,...,6
    double xsum=0, ysum=0, x2sum=0, xysum=0;
    for(int i=0; i<n; i++) {
        double x = i+1;
        double y = production[i];
        xsum += x;
        ysum += y;
        x2sum += x*x;
        xysum += x*y;
    }
    double beta = (n*xysum - xsum*ysum) / (n*x2sum - xsum*xsum);
    double alpha = (ysum - beta*xsum) / n;
    out[0] = alpha;
    out[1] = beta;
    out[2] = alpha + beta;
}
```

# Python-Code

```
# Code by GVV Sharma
# Modified for Problem Solution
# Released under GNU GPL
# Calculating area enclosed between curves
# call.py
import ctypes
import numpy as np

lib = ctypes.CDLL('./libregression.so')

get_data = lib.get_data
get_data.argtypes = [
    ctypes.POINTER(ctypes.c_double),
    ctypes.POINTER(ctypes.c_double),
    ctypes.POINTER(ctypes.c_int)
]
```

# Python-Code

```python
get_data.restype = None

years = np.zeros(6, dtype=np.double)
prod = np.zeros(6, dtype=np.double)
n = ctypes.c_int()
get_data(
    years.ctypes.data_as(ctypes.POINTER(ctypes.c_double)),
    prod.ctypes.data_as(ctypes.POINTER(ctypes.c_double)),
    ctypes.byref(n)
)

print('Years:', years[:n.value])
print('Production:', prod[:n.value])
```

# Python-Code

```
get_stats = lib.get_stats
stats = np.zeros(3, dtype=np.double)
get_stats(
    years.ctypes.data_as(ctypes.POINTER(ctypes.c_double)),
    prod.ctypes.data_as(ctypes.POINTER(ctypes.c_double)),
    n,
    stats.ctypes.data_as(ctypes.POINTER(ctypes.c_double))
)
print('Alpha:', stats[0])
print('Beta:', stats[1])
print('Alpha + Beta:', stats[2])
```

# Python-Code

```
import numpy as np
import matplotlib.pyplot as plt
import ctypes

# Load compiled .so file
lib = ctypes.CDLL('./libregression.so')

get_data = lib.get_data
get_data.argtypes = [
    ctypes.POINTER(ctypes.c_double),
    ctypes.POINTER(ctypes.c_double),
    ctypes.POINTER(ctypes.c_int)
]
get_data.restype = None

years = np.zeros(6, dtype=np.double)
prod = np.zeros(6, dtype=np.double)
n = ctypes.c_int()
```

# Python-Code

```python
get_data(
    years.ctypes.data_as(ctypes.POINTER(ctypes.c_double)),
    prod.ctypes.data_as(ctypes.POINTER(ctypes.c_double)),
    ctypes.byref(n)
)
x = np.arange(1, n.value + 1)
y = prod

# Get statistics using .so file
get_stats = lib.get_stats
stats = np.zeros(3, dtype=np.double)
get_stats(
    years.ctypes.data_as(ctypes.POINTER(ctypes.c_double)),
    prod.ctypes.data_as(ctypes.POINTER(ctypes.c_double)),
    n,
    stats.ctypes.data_as(ctypes.POINTER(ctypes.c_double))
)
alpha, beta, alpha_plus_beta = stats
```

# Python-Code

```
# Plot scatter and regression (bar for production, points for prediction)
plt.figure(figsize=(8,5))
plt.bar(years, prod, color='lightblue', label='Production')
plt.scatter(years, prod, color='black', zorder=2, label='Data Points')
reg_line = alpha + beta * (x-1)
plt.plot(years, reg_line, color='red', linewidth=2, label='Regression')
plt.xlabel('Year')
plt.ylabel('Production (lakh tonnes)')
plt.title(f'Regression Bar Graph & Line: $\\alpha+\\beta$={alpha_plus_beta:.2f}')
plt.legend()
plt.tight_layout()
plt.show()
```